Regression Metrics for Machine Learning

By Johannes Simatupang, MKom,COBIT5 (April 2021)

COMP6577 - Machine Learning - LB01 >>> Bina Nusantara University

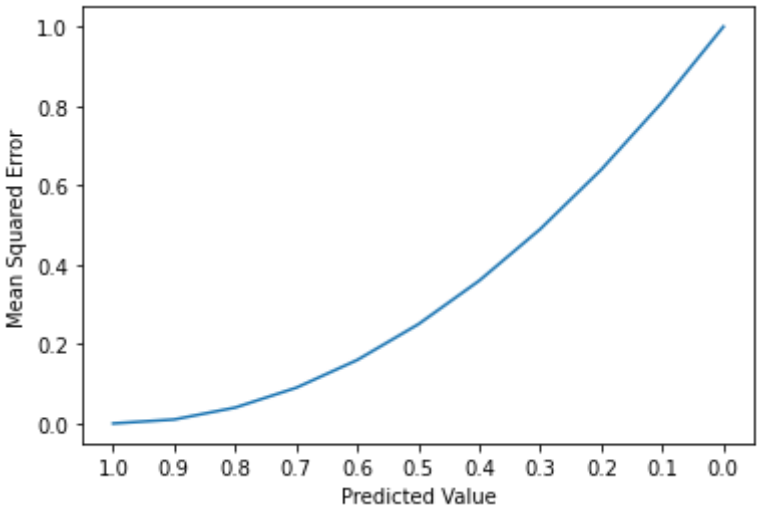
There are three error metrics that are commonly used for evaluating and reporting the performance of a regression model; they are:

Mean Squared Error (MSE). Root Mean Squared Error (RMSE). Mean Absolute Error (MAE)

In [1]:

```
# example of increase in mean squared error
from matplotlib import pyplot
from sklearn.metrics import mean_squared_error
# real value
expected = [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0]
# predicted value
predicted = [1.0, 0.9, 0.8, 0.7, 0.6, 0.5, 0.4, 0.3, 0.2, 0.1, 0.0]
# calculate errors
errors = list()
for i in range(len(expected)):
    # calculate error
    err = (expected[i] - predicted[i])**2
    # store error
    errors.append(err)
    # report error
    print('>%.1f, %.1f = %.3f' % (expected[i], predicted[i], err))
# plot errors
pyplot.plot(errors)
pyplot.xticks(ticks=[i for i in range(len(errors))], labels=predicted)
pyplot.xlabel('Predicted Value')
pyplot.ylabel('Mean Squared Error')
pyplot.show()
```

```
>1.0, 1.0 = 0.000
>1.0, 0.9 = 0.010
>1.0, 0.8 = 0.040
>1.0, 0.7 = 0.090
>1.0, 0.6 = 0.160
>1.0, 0.5 = 0.250
>1.0, 0.4 = 0.360
>1.0, 0.3 = 0.490
>1.0, 0.2 = 0.640
>1.0, 0.1 = 0.810
>1.0, 0.0 = 1.000
```



In [2]:

```
# example of calculate the mean squared error
from sklearn.metrics import mean_squared_error
# real value
expected = [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0]
# predicted value
predicted = [1.0, 0.9, 0.8, 0.7, 0.6, 0.5, 0.4, 0.3, 0.2, 0.1, 0.0]
# calculate errors
errors = mean_squared_error(expected, predicted)
# report error
print(errors)
```

```
0.35000000000000003
```

In [3]:

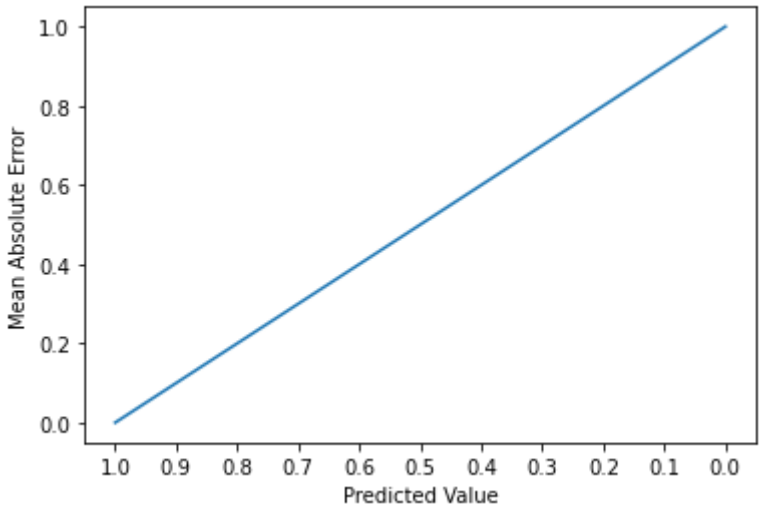
```
# example of calculate the root mean squared error
from sklearn.metrics import mean_squared_error
# real value
expected = [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0]
# predicted value
predicted = [1.0, 0.9, 0.8, 0.7, 0.6, 0.5, 0.4, 0.3, 0.2, 0.1, 0.0]
# calculate errors
errors = mean_squared_error(expected, predicted, squared=False)
# report error
print(errors)
```

```
0.5916079783099616
```

In [4]:

```
# plot of the increase of mean absolute error with prediction error
from matplotlib import pyplot
from sklearn.metrics import mean_squared_error
# real value
expected = [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0]
# predicted value
predicted = [1.0, 0.9, 0.8, 0.7, 0.6, 0.5, 0.4, 0.3, 0.2, 0.1, 0.0]
# calculate errors
errors = list()
for i in range(len(expected)):
    # calculate error
    err = abs((expected[i] - predicted[i]))
    # store error
    errors.append(err)
    # report error
    print('>%.1f, %.1f = %.3f' % (expected[i], predicted[i], err))
# plot errors
pyplot.plot(errors)
pyplot.xticks(ticks=[i for i in range(len(errors))], labels=predicted)
pyplot.xlabel('Predicted Value')
pyplot.ylabel('Mean Absolute Error')
pyplot.show()
```

```
>1.0, 1.0 = 0.000
>1.0, 0.9 = 0.100
>1.0, 0.8 = 0.200
>1.0, 0.7 = 0.300
>1.0, 0.6 = 0.400
>1.0, 0.5 = 0.500
>1.0, 0.4 = 0.600
>1.0, 0.3 = 0.700
>1.0, 0.2 = 0.800
>1.0, 0.1 = 0.900
>1.0, 0.0 = 1.000
```



In [5]:

```
# example of calculate the mean absolute error
from sklearn.metrics import mean_absolute_error
# real value
expected = [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0]
# predicted value
predicted = [1.0, 0.9, 0.8, 0.7, 0.6, 0.5, 0.4, 0.3, 0.2, 0.1, 0.0]
# calculate errors
errors = mean_absolute_error(expected, predicted)
# report error
print(errors)
```

```
0.5
```

source : https://machinelearningmastery.com/regression-metrics-for-machine-learning/